

Design Verification Plan & Report - Power-Tool Battery (case t3_r3)

<i>VBF-T3R3-DVPR-01. Virtual test version: all results from simulation protocols. Uncovered conditions honestly written "N/A". No numbers from memory.</i>

Item	Condition	Result	Determination	Source
1C discharge capacity	run-pyamm --protocol 1C_discharge (DFN)	4.3176 Ah	PASS vs >= 2.0 Ah	r5_v12_1c_dfn.json:capacity_ah
5C discharge capacity retention	run-pyamm --protocol 5C_discharge (DFN) / 1C	98.62% (4.2581 Ah / 4.3176 Ah)	PASS vs >= 95%	r5_v12_derived.json:retention_5c
4C fast-charge temperature rise	run-pyamm --protocol 4C_charge_45C --thermal lumped (DFN)	T_max 329.75 K (56.60 C)	PASS vs <= 333.15 K	r5_v12_4c_dfn.json:T_max_K
4C fast-charge lithium plating	same protocol + --plating; anode potential < 0 V	anode min +41.7 mV	PASS (no plating)	r5_v12_4c_dfn.json:anode_potential_v
5C discharge temperature	run-pyamm --protocol 5C_discharge --thermal lumped (DFN)	T_max 319.57 K (46.42 C)	PASS vs <= 333.15 K (informational)	r5_v12_5c_dfn.json:T_max_K
Power density	calc-energy contract formula $V_{OC2}/(4*DCR)/mass$	181,557 W/kg	PASS vs >= 4,000 W/kg	r5_v12_energy.json:power_density_w_kg
Voltage window	parameter set upper/lower cut-offs	2.5-4.2 V	PASS (design basis)	pybamm Chen2020 parameter set
Nail penetration	physical abuse test	-	N/A (beyond pure simulation boundary, requires physical experiment)	-
Overcharge to thermal runaway	physical safety test	-	N/A (beyond pure simulation boundary, requires physical experiment)	-
Crush / drop	mechanical abuse test	-	N/A (beyond pure simulation boundary, requires physical experiment)	-
Cycle life	aging protocol (SEI growth)	-	N/A (not simulated; no durability objective in task text)	-
Rate-pulse DC internal resistance	pulse protocol	-	N/A (not simulated; DCR estimate 1.059 mOhm at 1C mid-point for reference)	r5_v12_energy.json:dcr_ohm

Conclusion

All five entry-0 criteria verified PASS at DFN judge grade (round 5). Uncovered items: nail penetration, overcharge-to-runaway, crush/drop, cycle life, rate-pulse DCIR - all physical-experiment or aging-model territory, cited directly as paper limitations.